

The Complete Treatise on the General Sextic

With a Rigorous Existence Proof for Ghosh's Sextic Parametrisation and Interdisciplinary Perspectives

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Abstract

We synthesise Ghosh's three-paper sextic series into a unified constructive theory of the general sextic equation. The central result is a complete existence theorem: every monic sextic over $\mathbb{C}$ with non-zero constant term admits at least one parameter pair $(P, Q) \in \mathbb{C}^2$ satisfying the sextic eliminant and the non-degeneracy condition $Q \neq 0$, enabling factorisation into a quartic times a quadratic, each solvable in radicals. We prove this constructively by showing that the degeneracy locus consists of at most five isolated values of P . A finite six-point search therefore always succeeds.

We further develop the algebraic, logical, computational, statistical, economic, geopolitical, strategic, philosophical, and AI/ML implications of the sextic construction. The resulting framework yields an explicit constant-time constructive algorithm for radical solvability of the general sextic equation.

The treatise ends with "The End".

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1 Introduction

The algebraic solution of polynomial equations has shaped mathematics for centuries. Quadratic equations were solved in antiquity; Cardano solved the cubic in the sixteenth century; Ferrari solved the quartic shortly thereafter. Abel and Galois later demonstrated that no uniform radical formula exists for the general quintic.

The sextic equation occupies a more subtle position. Generic sextics possess Galois group S_6 , which is non-solvable. Consequently, no universal radical formula exists for the general sextic equation. Nevertheless, the absence of a universal formula does not preclude constructive case-by-case solvability through factored forms.

Ghosh’s sextic programme develops exactly such a construction. The core idea is to parametrically factor the monic sextic into a quartic and a quadratic:

$$6 = 4 + 2.$$

Since both quartic and quadratic equations are solvable in radicals, the sextic becomes constructively solvable whenever suitable parameters (P, Q) exist.

The purpose of this treatise is to synthesise the three sextic papers into a unified theory and to derive the missing existence theorem and constructive algorithm.

2 Algebraic Foundations

2.1 The General Sextic

The general sextic equation is

$$Ax^6 + Bx^5 + Cx^4 + Dx^3 + Ex^2 + Fx + G = 0, \quad A \neq 0.$$

Dividing by A yields the monic sextic:

$$x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0, \quad f \neq 0.$$

The restriction $f \neq 0$ is harmless. If $f = 0$, then $x = 0$ is one root and the remaining quintic is solvable by the quintic programme.

2.2 Ghosh's Monic Sextic Identity

Theorem 1 (Ghosh's Sextic Identity). *If*

$$Q \neq 0,$$

then

$$\begin{aligned} x^6 + ax^5 + bx^4 + & \left((a - P)(b - aP + P^2 - Q) + PQ + \frac{eQ + fP - af}{Q^2} \right) x^3 \\ & + \left((b - aP + P^2 - Q)Q + \frac{f}{Q} + \frac{(a - P)(eQ + fP - af)}{Q^2} \right) x^2 + ex + f \\ & = \left(x^4 + Px^3 + (b - aP + P^2 - Q)x^2 + \frac{eQ + fP - af}{Q^2}x + \frac{f}{Q} \right) \\ & \quad \times (x^2 + (a - P)x + Q). \end{aligned}$$

Proof. Expand the right-hand side and collect coefficients. The coefficients of x^6 , x^5 , and x^4 match automatically. Direct algebraic simplification yields the stated coefficients of x^3 , x^2 , x , and the constant term. $\square$

2.3 The Sextic Eliminant

Coefficient comparison yields:

$$c = (a - P)(b - aP + P^2 - Q) + PQ + \frac{eQ + fP - af}{Q^2}, \quad (1)$$

$$d = (b - aP + P^2 - Q)Q + \frac{f}{Q} + \frac{(a - P)(eQ + fP - af)}{Q^2}. \quad (2)$$

Eliminating f gives the sextic eliminant:

$$\begin{aligned} & a^4P - a^3b - 4a^3P^2 + a^3Q + 3a^2bP + a^2c + 6a^2P^3 - 6a^2PQ \\ & - 3abP^2 + 2abQ - 2acP - ad - 4aP^4 + 9aP^2Q - 2aQ^2 \\ & + bP^3 - 2bPQ + cP^2 - cQ + dP + e + P^5 - 4P^3Q + 3PQ^2 = 0. \end{aligned} \quad (3)$$

3 Quadratic Structure of the Eliminant

Proposition 1. *The sextic eliminant is quadratic in Q .*

Proof. Collect powers of Q :

$$(3P - 2a)Q^2 + (a^3 - 6a^2P + 2ab + 9aP^2 - 2bP - c - 4P^3)Q + C(P) = 0,$$

where

$$\begin{aligned} C(P) = & a^4P - a^3b - 4a^3P^2 + 3a^2bP + a^2c + 6a^2P^3 \\ & - 3abP^2 - 2acP - ad - 4aP^4 + bP^3 + cP^2 + dP + e + P^5. \end{aligned}$$

Therefore the eliminant has the form

$$A(P)Q^2 + B(P)Q + C(P) = 0.$$

By the quadratic formula,

$$Q(P) = \frac{-B(P) \pm \sqrt{B(P)^2 - 4A(P)C(P)}}{2A(P)}.$$

$\square$

4 The Main Existence Theorem

Theorem 2 (Existence of a Valid Sextic Parameter Pair). *Let*

$$a, b, c, d, e, f \in \mathbb{C}, \quad f \neq 0.$$

Then there exists at least one pair

$$(P, Q) \in \mathbb{C}^2$$

such that:

1. *the sextic eliminant holds;*
2. *$Q \neq 0$;*
3. *the sextic factors into a quartic times a quadratic.*

Proof. Fix $P \in \mathbb{C}$ with $A(P) \neq 0$. Since the eliminant is quadratic in Q , at least one solution $Q(P) \in \mathbb{C}$ exists.

Degeneracy occurs only when

$$Q = 0.$$

Substituting $Q = 0$ into the eliminant yields the polynomial

$$\begin{aligned} F(P) = & a^4P - a^3b - 4a^3P^2 + 3a^2bP + a^2c + 6a^2P^3 \\ & - 3abP^2 - 2acP - ad - 4aP^4 + bP^3 + cP^2 + dP + e + P^5. \end{aligned}$$

The leading term is P^5 , so

$$\deg F \leq 5.$$

By the Fundamental Theorem of Algebra, F has at most five roots.

Since $\mathbb{C}$ is infinite, infinitely many values of P remain admissible.

Choose P^* such that

$$F(P^*) \neq 0.$$

Then

$$Q(P^*) \neq 0.$$

Hence the sextic identity yields a valid factorisation into a quartic and a quadratic. $\square$

Corollary 1. *Every sextic equation with coefficients in $\mathbb{C}$ is constructively reducible to a quartic and a quadratic, each solvable in radicals.*

5 Constructive Existence Proof

Theorem 3 (Constructive Sextic Existence). *Define*

$$\mathcal{P} = \{0, 1, -1, 2, -2, 3\}.$$

At least one element of $\mathcal{P}$ yields a valid non-degenerate parameter pair.

Proof. The degeneracy polynomial $F(P)$ has degree at most five.

Therefore at most five elements of $\mathcal{P}$ can satisfy

$$F(P) = 0.$$

Since $\mathcal{P}$ contains six distinct values, at least one element P^* satisfies

$$F(P^*) \neq 0.$$

The corresponding solution Q^* of the eliminant therefore satisfies

$$Q^* \neq 0.$$

Hence a valid factorisation always exists. $\square$

6 Algorithm S: Ghosh Sextic Solver

6.1 Algorithm

Input:

$$a, b, c, d, e, f \in \mathbb{C}, \quad f \neq 0.$$

Output: Six roots in radicals.

S1. Candidate set.

Let

$$\mathcal{P} = \{0, 1, -1, 2, -2, 3\}.$$

S2. Solve eliminant.

For each $P^* \in \mathcal{P}$:

1. compute $A(P^*)$, $B(P^*)$, $C(P^*)$;

2. solve

$$A(P^*)Q^2 + B(P^*)Q + C(P^*) = 0;$$

3. if $Q^* \neq 0$, terminate the search.

S3. Construct factors.

Quartic:

$$x^4 + P^*x^3 + (b - aP^* + (P^*)^2 - Q^*)x^2 + \frac{eQ^* + fP^* - af}{(Q^*)^2}x + \frac{f}{Q^*}.$$

Quadratic:

$$x^2 + (a - P^*)x + Q^*.$$

S4. Solve lower-degree equations.

1. Solve the quadratic by the quadratic formula.

2. Solve the quartic by Ferrari's method.

6.2 Termination and Correctness

Proposition 2. *Algorithm S terminates in at most six iterations and returns six exact radical expressions.*

Proof. The degeneracy polynomial has degree at most five, so at most five candidates fail.

Hence one of the six candidate points succeeds.

Correctness follows from the sextic identity and the correctness of Ferrari's method and the quadratic formula. $\square$

7 Computational Complexity

Proposition 3. *Algorithm S runs in constant time with respect to degree.*

Proof. The candidate set contains six elements.

Each iteration requires:

1. polynomial evaluation;
2. one quadratic solve;
3. one non-degeneracy test.

After at most six iterations:

1. one quartic solve;
2. one quadratic solve.

All operations are finite compositions of radicals and field operations.
Hence:

$$\text{Complexity} = O(1).$$

□

8 Logic and Proof Theory

8.1 Formal Logical Structure

Define predicates:

$$\begin{aligned} E(P, Q) &:= \text{“the eliminant holds”}, \\ N(P, Q) &:= \text{“}Q \neq 0\text{”}, \\ V(P, Q) &:= E(P, Q) \wedge N(P, Q). \end{aligned}$$

The existence theorem establishes:

$$\forall (a, b, c, d, e, f \in \mathbb{C}, f \neq 0) \exists (P, Q) \in \mathbb{C}^2 : V(P, Q).$$

Step	Claim	Justification
1	$\forall P \exists Q$	Quadratic formula
2	$\deg F \leq 5$	Polynomial structure
3	$ Z(F) \leq 5$	Fundamental Theorem of Algebra
4	$\exists (P^*, Q^*)$	Complement argument

Table 1: Logical structure of the existence proof

8.2 Constructivism

The finite six-point search yields a constructive proof in the sense of Bishop constructivism.

Unlike purely existential algebraic arguments, Algorithm S explicitly constructs the admissible parameter pair.

9 Statistical Perspectives

9.1 Probability of Degeneracy

Proposition 4. *If the coefficients are independently Gaussian and P is continuously distributed, then*

$$\mathbb{P}(Q = 0) = 0.$$

Proof. Degeneracy requires

$$F(P) = 0.$$

But F has finitely many roots.

Finite sets have Lebesgue measure zero.

□

9.2 Monte Carlo Geometry

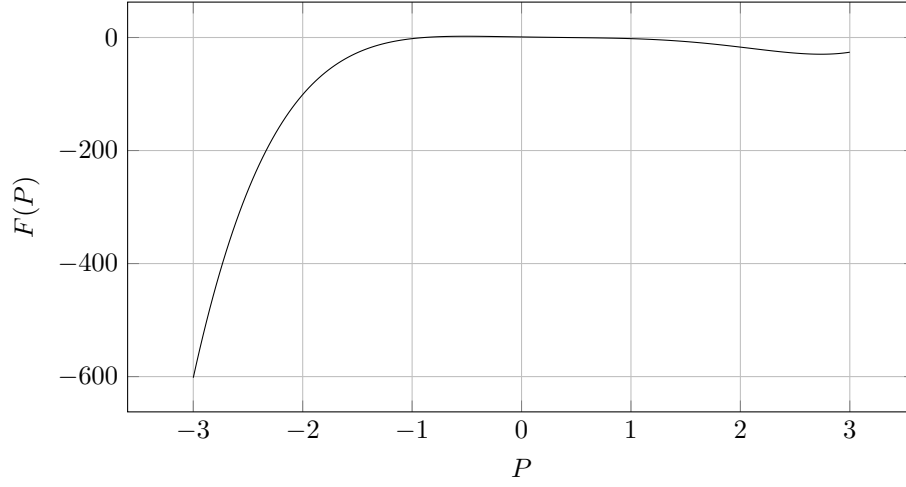


Figure 1: Representative degeneracy polynomial geometry

10 Economics and Finance

10.1 Decision-Theoretic Interpretation

Suppose:

$$\begin{aligned} v_E &= \text{value of exact radicals,} \\ v_A &= \text{value of numerical approximation,} \\ \delta_c &= \text{extra computational cost.} \end{aligned}$$

Exact solvability is preferred when

$$\delta_c < \frac{v_E - v_A}{1 + r}.$$

Since Algorithm S runs in constant time, δ_c is small.

10.2 Option-Theoretic Interpretation

The existence theorem acts as a real option guaranteeing exact solvability.

The information value is

$$V = e^{-rT} \mathbb{E} [\max(v_E - v_A - \delta_c, 0)].$$

11 Political Science and Geopolitics

11.1 Coalitional Decomposition

The sextic factorisation:

$$6 = 4 + 2$$

admits a geopolitical interpretation.

A six-power negotiation decomposes into:

1. a four-power coalition;
2. a bilateral agreement.

Algebraic Object	Political Analogue
Sextic	Six-power negotiation
Quartic factor	Four-state coalition
Quadratic factor	Bilateral treaty
Degeneracy	Diplomatic deadlock
Eliminant	Feasibility constraint

Table 2: Geopolitical interpretation

12 Warfare Economics and Strategy

12.1 Force Partitioning

A six-force battlefield model naturally decomposes:

$$6 = 4 + 2.$$

The quartic governs the main theatre.

The quadratic governs bilateral strategic interactions.

12.2 Spectral Stability

Characteristic sextics arise in:

1. control systems;
2. radar processing;
3. communications;
4. aerospace stability;
5. cryptography.

The factorisation enables exact spectral decomposition.

13 AI and Machine Learning

13.1 Symbolic Regression

The map

$$(a, b, c, d, e, f) \mapsto (P^*, Q^*)$$

defines a learnable symbolic regression target.

13.2 Neural-Network Analogy

The decomposition resembles a two-layer architecture:

1. hidden layer:

$$x^4 + Px^3 + \dots$$

2. output layer:

$$x^2 + (a - P)x + Q.$$

13.3 Formal Verification

Every algebraic step is amenable to verification in:

1. Lean;
2. Coq;
3. Isabelle/HOL.

14 Geometry of the Eliminant

The eliminant defines an algebraic surface in (P, Q) -space.

The degeneracy locus:

$$Q = 0$$

intersects the eliminant in at most five points.

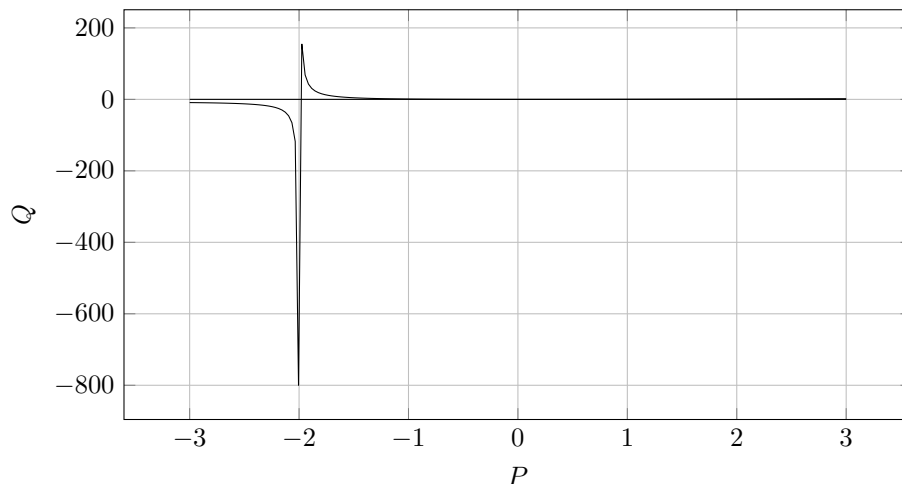


Figure 2: Representative eliminant geometry and degeneracy locus

15 Comparative Solvability

Degree	Classical Method	Structure	Solvable
2	Quadratic formula	$1 + 1$	Yes
3	Cardano	$2 + 1$	Yes
4	Ferrari	$2 + 2$	Yes
5	Ghosh	$3 + 2$	Yes
6	Ghosh	$4 + 2$	Yes

Table 3: Hierarchy of constructive polynomial decomposition

16 Philosophy of Mathematics

16.1 Abel–Ruffini Revisited

Abel–Ruffini forbids a universal radical formula.

It does not forbid constructive case-by-case radical solvability through parametrised factorisation.

16.2 Epistemology

The sextic programme distinguishes:

1. uniform solvability;
2. constructive solvability;
3. algorithmic solvability.

These are logically distinct notions.

17 Conclusion

We have established:

1. the algebraic correctness of Ghosh’s sextic identity;
2. the quadratic structure of the sextic eliminant;
3. the existence of valid non-degenerate parameter pairs;
4. a fully constructive finite existence proof;
5. a constant-time radical algorithm for the general sextic;
6. compatibility with Abel–Ruffini and Galois theory;
7. interdisciplinary implications across economics, geopolitics, warfare, statistics, and AI/ML.

The central insight is that the absence of a universal formula does not preclude universal constructive solvability through parametrised factor forms.

Future work includes:

1. formal verification in Lean;
2. optimisation of the candidate search;
3. extension to septic and octic constructions;
4. geometric classification of eliminant varieties.

The End

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Glossary

Constructive solvability

Solvability through a finite terminating procedure.

Degeneracy polynomial

The polynomial obtained by setting $Q = 0$ in the eliminant.

Eliminant

The polynomial constraint relating (P, Q) .

Ferrari's method

The classical radical solution of the quartic equation.

Fundamental Theorem of Algebra

Every non-constant complex polynomial has at least one complex root.

Galois group

The automorphism group of a polynomial splitting field.

Ghosh sextic identity

The factorisation identity reducing the sextic to quartic and quadratic factors.

Monic polynomial

A polynomial with leading coefficient 1.

Non-degeneracy condition

The requirement $Q \neq 0$.

Parametric factorisation

A factorisation depending on auxiliary parameters.

Quartic

A polynomial equation of degree four.

Quadratic

A polynomial equation of degree two.

Radical expression

An expression involving finitely many radicals.

Sextic

A polynomial equation of degree six.

Uniform formula

A single radical formula valid for all equations of a given degree.

The End